

GOLD-STANDARD INTEGRATED HEMATOPATHOLOGY REPORT

AML Integrated Reporting Workshop — Case 01 — pathologist-authored reference for scoring model output. Fictional patient and data, created for teaching.

1. PATIENT AND SPECIMEN IDENTIFICATION

Jonathan R. Doe, MRN-4471902, 58-year-old man. Specimen: bone marrow, left posterior iliac crest, aspirate and core biopsy, collected 2026-04-22. Ordering provider: Anjali R. Patel, MD, Hematology/Oncology.

2. COMPONENT STUDIES REVIEWED

Component	Accession	Performing laboratory	Reported
Bone marrow morphology	S26-BM-01187	Riverbend Regional Medical Center	2026-04-24
Flow cytometry	FC-26-003942	Riverbend Regional Medical Center	2026-04-23
Cytogenetics + FISH	CG-26-00871	Riverbend Regional Medical Center	2026-04-29
Molecular NGS (54-gene)	MMD-26-NGS-22150	Meridian Molecular Diagnostics (reference lab)	2026-05-04

3. CLINICAL CONTEXT

Three weeks of progressive fatigue, exertional dyspnea, gingival bleeding, and lower-extremity ecchymoses. Admission CBC: WBC 38.4 x10⁹/L, hemoglobin 7.9 g/dL, platelets 31 x10⁹/L. Peripheral smear with 41% circulating blasts. No prior hematologic disease and no prior cytotoxic therapy or radiation. Marrow performed to evaluate for acute leukemia.

4. MORPHOLOGY SUMMARY

Markedly hypercellular marrow (90-95%) with extensive replacement by immature mononuclear cells. Manual 500-cell aspirate differential: blasts 18%, with promonocytes enumerated separately at 4%. Core biopsy shows blast-equivalent infiltration well in excess of the aspirate figure. A subset of blasts has folded nuclei and monocytoid cytoplasm, but morphology stopped short of committing to monocytic differentiation. No definite Auer rods. Cytochemistry: MPO positive in a minority of blasts; non-specific esterase positive with partial fluoride inhibition, favoring a monocytic component. Reticulin MF-1, no ring sideroblasts. Morphologic diagnosis: hypercellular marrow with acute myeloid leukemia, probable monocytic differentiation, pending ancillary studies.

5. FLOW CYTOMETRY SUMMARY

An abnormal myeloid blast population is 22% of total nucleated cells, expressing CD34, CD117, HLA-DR, CD13, CD33, CD38, and dim cytoplasmic MPO, with aberrant CD56 and partial aberrant CD7. There is a markedly expanded, left-shifted monocytic compartment (maturing monocytes 19% plus immature monocytic/promonocytic forms 11%, together about 30% of total nucleated cells) with bright uniform CD64, CD14 on the maturing fraction, CD11b, CD11c, CD36, HLA-DR, and dim CD4. Flow interpretation: acute myeloid leukemia with monocytic differentiation; aberrant CD56/CD7 suitable for measurable residual disease tracking.

6. CYTOGENETICS SUMMARY

Conventional G-banded karyotype: 46,XY[20], a normal male complement. Interphase FISH AML panel negative for RUNX1-RUNX1T1, CBFB, PML-RARA, KMT2A, MECOM, del(5q), monosomy 7/del(7q), and del(17p), all within laboratory cutoffs. The case is cytogenetically normal. No recurrent AML cytogenetic abnormality by either method.

7. MOLECULAR SUMMARY

Targeted 54-gene myeloid NGS detected three variants. NPM1 NM_002520.7:c.860_863dup p.(Trp288CysfsTer12), a type A frameshift insertion, VAF 39%, Tier I. FLT3 internal tandem duplication, exon 14, VAF 31%, ITD/WT allelic ratio 0.62, Tier I. DNMT3A NM_022552.5:c.2645G>A p.(Arg882His), R882 hotspot missense, VAF 44%, Tier II. No reportable variant in CEBPA, RUNX1, TP53, ASXL1, IDH1, IDH2, or the remaining panel genes. NPM1 is a disease-defining abnormality. FLT3-ITD is prognostically adverse and FLT3-inhibitor targetable. DNMT3A R882H is prognostically relevant but does not by itself define or change the disease category.

8. INTEGRATED INTERPRETATION

1. Morphology, flow cytometry, cytogenetics, and molecular studies on a single bone marrow specimen together establish acute myeloid leukemia.
2. The two blast estimates differ: the manual aspirate differential gave 18% while flow cytometry gated 22% of total nucleated cells as abnormal blasts.
3. This difference is expected between a manual smear count and a gated flow estimate, is compounded by aspirate hemodilution and by the difficulty of separating promonocytes from myeloid blasts on smears, and the core biopsy showed blast-equivalent infiltration well above either figure.
4. More importantly, the blast percentage is not the deciding factor in this case, because a defining genetic abnormality is present and the 20% threshold does not apply to it.
5. Morphology favored but did not commit to monocytic differentiation; flow cytometry resolved this by demonstrating a large, left-shifted monocytic compartment with bright CD64, CD14 on the maturing fraction, CD11b, CD11c, and CD36.
6. Conventional karyotype and the FISH panel were both normal, placing the case in the cytogenetically normal group and excluding the recurrent AML rearrangements.
7. The decisive findings came from molecular testing alone: mutated NPM1, an FLT3 internal tandem duplication, and a DNMT3A R882H variant, none of which are visible by morphology, flow, or cytogenetics.
8. Mutated NPM1 is the disease-defining genetic abnormality and sets the WHO and ICC category.
9. The FLT3-ITD does not change the entity but is prognostically adverse and identifies the patient as a candidate for FLT3-inhibitor therapy, and the reported allelic ratio of 0.62 supports treatment decisions.
10. The DNMT3A R882H variant is commonly co-mutated with NPM1 and FLT3-ITD and is reported for prognostic completeness; it does not define or change the disease category and is not part of the diagnostic line.

9. FINAL INTEGRATED DIAGNOSIS

D1. BONE MARROW, LEFT POSTERIOR ILIAC CREST, ASPIRATE AND CORE BIOPSY:

D2. ACUTE MYELOID LEUKEMIA WITH MUTATED NPM1, WITH MONOCYTIC DIFFERENTIATION.

D3. FLT3 INTERNAL TANDEM DUPLICATION PRESENT (ITD/WT ALLELIC RATIO 0.62); NORMAL KARYOTYPE (46,XY).

10. PROGNOSTIC AND PREDICTIVE NOTES

By current risk frameworks the combination of mutated NPM1 with a concurrent FLT3-ITD is generally not treated as favorable risk, and the FLT3-ITD allelic ratio is provided to support that assessment and FLT3-inhibitor selection. Aberrant blast CD56 and CD7 by flow give a measurable residual disease immunophenotype. The DNMT3A R882H variant may carry additional prognostic weight and may represent a clonal hematopoiesis founder event; it is noted here, not in the diagnostic line.

11. LIMITATIONS AND PENDING OR RECOMMENDED STUDIES

The aspirate manual blast count is subject to hemodilution and to subjective promonocyte assignment. The NGS assay has an approximate 5% VAF limit of detection and does not reliably detect large structural rearrangements or copy-number changes; a negative result does not exclude a low-level variant. Karyotype resolution was 400-425 bphs. No further studies are required for classification; correlation with the treating hematology team for risk-adapted therapy is recommended.

PART B — EVIDENCE TRACE (sections 8 and 9)

Each interpretation and diagnosis sentence mapped to its supporting source(s) and basis. This is the scoring backbone: a model sentence with no traceable support, or support that the cited source does not actually contain, counts as a hallucination.

#	Sentence (abbreviated)	Source(s)	Basis
8.1	Four modalities together establish AML.	MORPH, FLOW, CYTO, MOLEC	concordance
8.2	Blast estimates differ, 18% manual vs 22% flow.	MORPH, FLOW	discordance_resolution
8.3	Difference expected; method variance, hemodilution, core above both.	MORPH, FLOW	discordance_resolution
8.4	Blast % not deciding factor; defining lesion present, 20% rule N/A.	MORPH, MOLEC	classification_rule
8.5	Flow resolved monocytic differentiation morphology hedged on.	MORPH, FLOW	discordance_resolution
8.6	Karyotype and FISH normal; cytogenetically normal group.	CYTO	direct_finding
8.7	Decisive findings molecular-only; invisible to other three.	MOLEC	single_source_finding
8.8	Mutated NPM1 is the disease-defining abnormality.	MOLEC	classification_rule
8.9	FLT3-ITD adverse and targetable; allelic ratio 0.62.	MOLEC	direct_finding
8.10	DNMT3A R882H prognostic only; not classifying.	MOLEC	direct_finding
9.D1	Specimen identification line.	MORPH	direct_finding
9.D2	AML with mutated NPM1, with monocytic differentiation.	MOLEC (NPM1) + FLOW (monocytic)	classification_rule
9.D3	FLT3-ITD present; normal karyotype.	MOLEC, CYTO	direct_finding

INSTRUCTOR NOTES — WHAT THIS CASE IS TESTING

Discordance 1, blast count. 18% (morphology) vs 22% (flow). A model that picks one number silently, or that argues the case hinges on crossing 20%, has missed the point. Correct handling: acknowledge both, explain the expected variance, then note the threshold is moot because a defining lesion is present.

Discordance 2, lineage. Morphology hedges on monocytic differentiation; flow proves it. A model should credit flow with resolving the hedge, not treat the two as contradictory.

Single-source finding. NPM1, FLT3-ITD, and DNMT3A exist only in the molecular report. Karyotype and FISH are normal. This is the case's central argument for integrated reporting and the model should say so explicitly.

Planted near-miss, DNMT3A R882H. Real, Tier II, prognostically relevant, but not classifying. A model that writes it into the diagnosis line or uses it to assign the entity has failed the in-its-lane test. It belongs in the molecular summary and prognostic notes only.

Scoring suggestion. Completeness (all four modalities represented), accuracy (diagnosis matches D2), discordance handling (both addressed explicitly), hallucination rate (every section 8-9 sentence traceable), and lane discipline (DNMT3A not in the dx). Run the same input three times to show consistency or its absence.

Reference authored for the AML Integrated Reporting Workshop. Fictional case. Not a real patient and not for clinical use.